

Reliability evaluation of MPPT based interleaved boost converter for PV system

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Abstract— The demand for power supply and depletion of the conventional energy sources are increasing drastically. So to overcome this problem, the best alternative power generation for conventional fossil fuel is Photovoltaic solar cell based system because of its advantage of pollution free and its availability in abundance with free of cost. In the MPPT based PV system the converters are the most sensitive part. Therefore to provide uninterrupted power supply without compromising the quality of power, reliability evaluation of interleaved boost converter becomes necessary. MATLAB/Simulink is used for the simulation studies and to determine the power losses of various components of the converter which is used in calculating the failure rates and reliability of the interleaved boost converter. Reliability studies of IBC have not been studied much. However there exists few literature in which reliability expression has been developed using Markov technique which is a more complex method as compare to Reliability Block Diagram (RBD). Therefore this paper proposes reliability modeling and reliability evaluation of Interleaved boost converter in MPPT based photo-voltaic system by using simple RBD method.

Keywords—Maximum Power Point Tracking (MPPT), Photovoltaic systems, Reliability, Failure rate, Reliability Block diagram(RBD), Interleaved Boost Converter (IBC).

I. INTRODUCTION

There are many devices and equipments around us which internally operate on direct current (DC). These devices are plugged into an alternating current (AC) outlet. For successful operation of these devices, AC to DC conversion becomes necessary which is not totally efficient and during this conversion the energy is being wasted. The demand of power supply and depletion of conventional energy sources are increasing drastically. So the use of renewable energy becomes only option to overcome that problem. Most of the newly constructed buildings are deploying solar energy sources that generate DC power. So, these converters are used to get a stabilized output voltage because the load is of DC type.

For off grid power generation PV energy based power system has emerged as the most popular alternative. PV system investment demands high initial cost and runs with almost negligible cost. Since this system primarily depends upon the solar irradiance, so connecting load directly to the panels will not be feasible. Because of different environment condition the deviation in generating power from this system arises. To minimize this effect and to enhance the generating energy from Photo Voltaic panels, Maximum Power Point

Tracking (MPPT) is used. Various MPPT methods have been evolved and put into practice [1].

In standard Photovoltaic system the Boost Converters are used both as power processing unit and Maximum Power point tracker. Reliability of PV system largely depends on the converters used such as Boost converter. If any of the component of the converter system fails, the overall system gets compromised which may lead to the failure of whole system. So reliability evaluation of these converters becomes must. Reliability study of these converters has very rarely done and there exists only a few papers in which complex method is used for reliability evaluation [2].

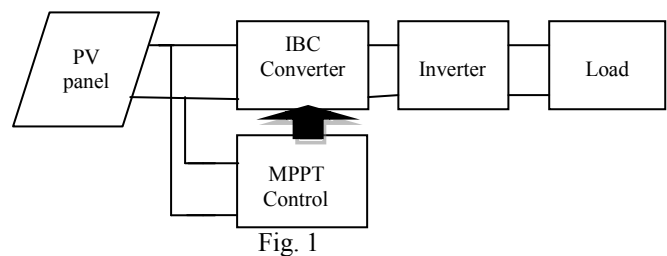


Fig. 1

Fig. 1 shows the block diagram for PV system with IBC converter.

A conventional boost converter steps up (boosts) a DC Voltage to a higher value. The converter's working depends upon the duty cycle and that is introduced from the maximum power point tracking according to Photovoltaic panel condition [3]. Additional benefit such as minimized ripple currents in input as well as output circuits is realized. Inductor ac loss also gets reduced as splitting of the output current into two paths.

This paper determines the reliability study of Photo Voltaic system based on MPPT with the implementation of IBC. The complete modeling of the proposed converter is done with the help of MATLAB/Simulink and for the reliability evaluation Reliability Block Diagram is developed.

This paper is organized as follows: section II gives the basic knowledge of PV Panel. Section III focuses on the modeling of IBC. Section IV and V provide the reliability concept, reliability modeling and calculation. Section VI presents the various simulation results and concluding remarks and future scope are provided in section VII.

II. PHOTOVOLTAIC PANEL

Solar cells are built up of semiconductor that has ability of absorbing the irradiance and temperature from the sun and converting into electrical power [4]. The connection of these modules can be done in two ways i.e. series and/or parallel. Series connections are made for obtaining large output voltage while parallel connection is for large output current. The following equation defines the relationship between voltage-current [5].

$$I = I_{pv} - I_s \left(\exp \frac{q(V+R_s I)}{N_s k T a} - 1 \right) - \frac{(V+R_s I)}{R_p} \quad (1)$$

Here:

- T : Temperature (K)
- I_{pv} : Photovoltaic Current (A)
- R_s : series resistance of cell (Ω)
- K : Boltzmann constant (1.38×10^{-23} J/K)
- N_s : Number of cells connected in series
- I_s : Saturation current
- Q : Electron charge (1.6×10^{-19} C)
- R_p : Resistance connected in parallel (Ω)
- a : Diode ideality constant

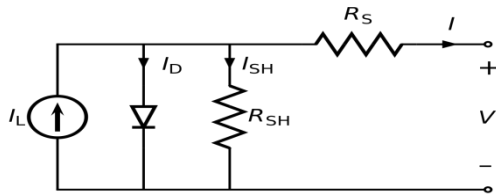


Fig. 2

Fig. 2 gives the idea of basic equivalent circuit diagram of a single diode Photovoltaic cell. An ideal PV cell may be designed by an independent current source having a parallel connected diode to it, but practically no cell is ideal. Therefore a shunt and a series resistance are connected in an ideal PV cell as shown in Fig. 2.

The output power of PV panel depends mainly on solar irradiance and temperature.

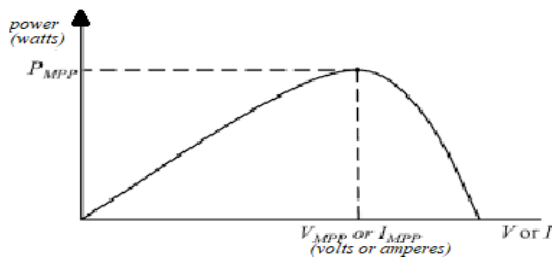


Fig. 3

Figure 3 shows power and voltage characteristic as a relation between power and voltage of a PV panel. From this characteristic it is clear that to get maximum power from the panel and to introduce an approximate duty cycle for firing the switches in the proposed converter, MPPT is needed.

The considered photovoltaic array in this proposed work consists of ten parallel strings with 748 cells in series in each string. The terminal voltage is given by [6,7]

$$V_{array} = 54.7 \log_e \left(\frac{I_{ph} - I_{array} + 0.005}{0.005} \right) - 3.74 I_{array} \quad (2)$$

Here, photocurrent is denoted by I_{ph} and it varies linearly to the solar insolation. ($I_{ph} = 8$ amp for 100% solar insolation of 1KW per square meter)

For power calculation multiplying above equation with I_{array} gives,

$$P_{array} = 54.7 I_{array} \log_e \left(\frac{I_{ph} - I_{array} + 0.005}{0.005} \right) - 3.74 I_{array}^2 \quad (3)$$

For maximum power,

$$\frac{dp_{array}}{dI_{array}} = 0,$$

After solving those relations, maximum power is evaluated for various insolation levels. The variation of maximum power to insolation levels is shown in the below figure (4). From the figure it is clear that maximum power is varying linearly with the insolation level.

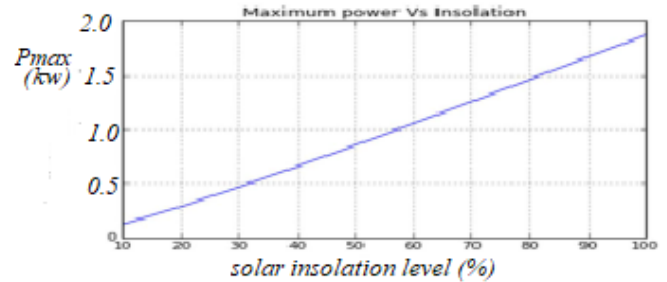


Fig. 4 P_{max} Vs Solar insolation level

This maximum power which is obtained by the solar panel has to be transferred via interleaved boost converter to the load. Hence evaluation of the reliability of this IBC becomes necessary which is used as MPPT system.

III. IBC MODELING

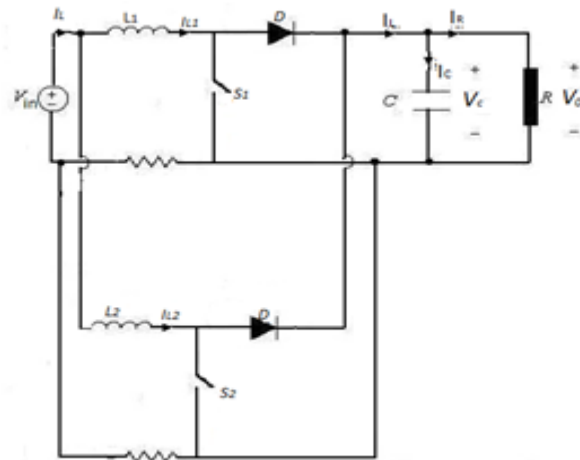


Fig. 5 Circuit diagram of IBC

The fig. 5 shows the simplified circuit model for interleaved boost converter where L_1 and L_2 are inductors of same value. V_{in} , D , C , R and V_o are input voltage, diode, capacitor, resistive load and output voltage respectively. Since both the parallel converters are identical, so inductor currents in both the converters are same.

So, $I_{L1} = I_{L2}$

For switch on condition,

$$I_{L1} = \int V_{in} u dt \quad (4)$$

$$I_{L2} = \frac{1}{L_2} \int V_{in} u dt \quad (5)$$

Also, $I_{L1} + I_{L2} = I_L$ and $I_c = I_L - I_R$

$$C \frac{dV_c}{dt} = -\frac{V_0}{R} \quad (6)$$

For switch off condition,

$$L_1 \frac{dI_{L1}}{dt} = V_{in} - V_0 \quad (7)$$

$$L_2 \frac{dI_{L2}}{dt} = V_{in} - V_0 \quad (8)$$

The duty cycle is described with the values of the output and the input voltage of the boost converter and shown in the below equation (9).

$$V_{output} = \frac{V_{input}}{1-\alpha} \quad (9)$$

Where, α is the duty cycle.

IV. RELIABILITY BASICS AND FAILURE RATE COMPUTATION

It is defined as the probability in which a system or a component, under given conditions and for a specific duration, will perform its objective without being failed.

Reliability depends mainly on the material's properties which are used in the device under different conditions such as tension, ambient and stress condition. The calculation of the actual failure rates of the components of the device becomes necessary for the reliability evaluation of any electronic device. The actual failure rates of electronic devices are generally constant and it is denoted by λ .

The MIL-HDBK 217 handbook [8] provides the basic failure rate λ_b of the components of the electronic devices. The actual failure rate is determined by using following relation [9],

$$\lambda = \lambda_b \left(\prod_{j=1}^n \pi_j \right)$$

Here 'n' is the total number of different factors affecting component's failure rate.

The overall failure rate of any device is calculated by addition of the failure rates of all the components of the device i.e.

$$\lambda_{total} = \sum_{i=1}^k \lambda_k$$

where 'k' denotes the number of components in the system.

And hence reliability R(t) is determined by,

$$R(t) = e^{-\lambda_{total} t} \quad (10)$$

The mean time to failure (MTTF) is calculated by,

$$MTTF = \int_0^{\infty} R(t) dt = \int_0^{\infty} e^{-\lambda_{total} t} dt = [\lambda_{total}]^{-1} \quad (11)$$

Power electronic devices mainly comprise of MOSFET, capacitors, inductors and diodes. The Table 1 shows the parameters that affect the failure rates of above mentioned components and also shows the relations for calculating the stress factor π_T which is also called temperature factor which depends on T_j , T_{HS} and T_a depending upon the component.

Where,

T_{HS} : Hot-spot Temperature

T_a : Ambient Temperature which is taken as 27°C.

T_j : Junction Temperature

Table 1
Factors affecting failure rates

MOSFET	$\lambda = \lambda_b \pi_T \pi_A \pi_Q \pi_E$
	$\pi_T = \exp \left(-1925 \left(\frac{1}{T_j + 273} - \frac{1}{298} \right) \right)$
Diode	$\lambda = \lambda_b \pi_T \pi_Q \pi_S \pi_E$
	$\pi_T = \exp \left(-3091 \left(\frac{1}{T_j + 273} - \frac{1}{298} \right) \right)$
Inductor	$\lambda = \lambda_b \pi_Q \pi_E \pi_T$
	$\pi_T = \exp \left(-1276.54 \left(\frac{1}{T_{HS} + 273} - \frac{1}{298} \right) \right)$
Capacitor	$\lambda = \lambda_b \pi_T \pi_E \pi_V \pi_Q \pi_C$
	$\pi_T = \exp \left(-4061.73 \left(\frac{1}{T_a + 273} - \frac{1}{298} \right) \right)$

Here, λ_b is basic failure rate, π_Q is quality factor, π_E is environment factor and π_A is the application factor. These factors are taken from Military handbook [8].

π_S is the stress factor and is used in determining the diode failure rate and this is affected the device voltage rating. The relationship is given by [10],

$$\pi_S = V_s^{2.43} \quad (12)$$

Also,

$$\pi_v = \left(\frac{V_s}{0.6} \right)^5 + 1 \quad (13)$$

Here v_s is the ratio of applied and rated voltage. π_C is the stress factor which is defined to calculate the failure rate of the capacitor and it depends on the value of capacitance.

$$\pi_C = C^{0.23} \text{ farad} \quad (14)$$

The stress factor π_T which is defined for all the components of the device is known as the Temperature Factor [9]. It is evaluated with the help of Table 1 for each of the components. The calculation of term T_j is very important and the methods for the calculations are given in reference [11]. In this paper the method used for finding the junction and hotspot temperature is by determining the power loss in each component of the converter. The following relationship is developed to calculate the device junction temperature.

$$T_j = T_c + \theta_{jc} P_D \quad (15)$$

$$\text{The case temperature, } T_c = T_a + \theta_{jA} P_D \quad (16)$$

θ_{jc} (°C/W) is used to denote the junction to case thermal resistance and has different value for different components and is taken from the datasheet of component. θ_{jA} is used to show the junction to ambient thermal resistance and is assumed to be 1°C/W.

For inductor's failure rate calculation, the hot-spot temperature is given by,

$$T_{HS} = T_a + 1.1 \Delta T \quad (17)$$

Here ΔT is the temperature measured for 5% of time.

For MOSFET power losses include the conduction loss i.e. P_c , gate drive loss i.e. P_{gd} and switching loss i.e. P_{sw} . The

below table 2 shows the power loss calculation for all the components of the device.

Table 2
Power loss of different components

MOSFET	$P_d = P_{sw} + P_c + P_{gd}$
	$P_{sw} = (E_{onM} + E_{offM})f_{sw}$
	$P_{gd} = C_{iss}V_{in}^2f_{sw}$
	$P_c = R_{DS(on)}I_{drms}^2$
DIODE	$P_d = V_{Do}I_{Favg} + R_D I_{FRMS}^2$
Inductor	$P_d = R_L I_L^2$

In the above Table 2,

- E_{onM} : turn-on energy loss of Switch
- f_{sw} : switching frequency
- E_{offM} : turn-off energy loss of Switch
- C_{iss} : input capacitance
- $R_{DS(on)}$: drain to source resistance
- I_{DRMS} : Mosfet on-state rms current
- I_{FRMS} : diode rms current
- I_{Favg} : diode average current
- V_{Do} : diode on-state zero-current voltage
- R_D : diode resistance
- R_L : inductor resistance
- I_L : inductor current

V. RELIABILITY MODELING AND EVALUATION

For the input voltage 150V and duty cycle of 0.5 the simulink model is developed and all the parameters are determined from the model.

From the Fig. (9-11),

For Diode,

$$V_n = 1200V, V_{do} = 1.65V \text{ and } \theta_{jc} = 1.6^\circ C/W.$$

$$R_D = 46.2m\Omega, I_{Favg} = 1.25A \text{ and } I_{FRMS}^2 = 3.125A$$

$$P_d = 2.207 W \text{ from table 2.}$$

$$T_j = 32.74^\circ C \text{ from equations (15,16).}$$

$$\pi_T = 2.123 \text{ from table 1.}$$

$$\lambda_b = 0.038, \pi_Q = 5.5, \pi_E = 1.$$

$$\pi_S = 0.0064, \lambda_{diode} = 3.24 \text{ failures}/10^7 \text{ hours}$$

For MOSFET,

$$\text{Power loss } P_d = 0.630 W$$

$$T_a = 27^\circ C$$

Hence, junction temperature

$$T_j = T_a + \theta_{JA} P_D + \theta_{JC} P_D = 28.638^\circ C$$

$$\text{So, } \lambda_{MOSFET} = 9.31 \text{ failures}/10^7 \text{ hours}$$

Similarly by using equation (17) and Table 1

$$\lambda_{inductor} = 4.96 \text{ failures}/10^{10} \text{ hours}$$

And by using equation (13,14) and Table 1

$$\lambda_{capacitor} = 0.386 \text{ failures}/10^7 \text{ hours}$$

So in a conventional boost converter if any of the above components fails whole system goes in failure state. So in the sense of reliability all those components are connected in series. M, L, D, C are the reliabilities of MOSFET, inductor, diode and capacitor respectively.



Fig. 6

Fig. 6 shows the reliability block diagram of conventional boost converter. Since its all components are connected in series. Hence its overall failure rate is,

$$\lambda = \lambda_{MOSFET} + \lambda_{diode} + \lambda_{inductor} + \lambda_{capacitor} = 12.936 \text{ failures}/10^7 \text{ hours}$$

$$\text{MTTF for conventional boost converter} = 88.246 \text{ years}$$

Therefore to improve the reliability, IBC is designed. In this topology one boost converter is added in the parallel hence its reliability block diagram is developed as below,

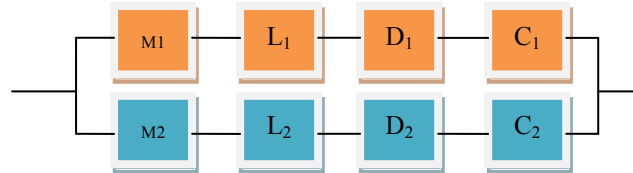


Fig. 7 RBD of IBC

Here M_1 and M_2 are the reliabilities of MOSFETs of IBC.

L_1 and L_2 are reliabilities of inductors of IBC.

Similarly D_1, D_2 and C_1, C_2 are the reliabilities of diodes and capacitors of the IBC respectively.

For parallel system, unreliability of the system is determined which is denoted by 'Q'.

$$Q_{IBC} = Q_1 Q_2 = (1 - R_1)(1 - R_2)$$

$$\text{Hence, } R_{IBC} = R_1 + R_2 - R_1 R_2 = e^{-\lambda_1} + e^{-\lambda_2} - e^{-(\lambda_1 + \lambda_2)}$$

$$\text{And, } \frac{1}{\lambda_{IBC}} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2} \quad (18)$$

Here subscript 1 and 2 are used for boost converters those are connected in parallel.

Since the components are identical in both the converter, so

$$\lambda_1 = \lambda_2 = \lambda$$

For interleaved boost converter,

$$\text{Failure rate } \frac{1}{\lambda_{IBC}} = \frac{2}{\lambda} - \frac{1}{2\lambda}$$

$$\lambda_{IBC} = 8.624 \text{ failures}/10^7 \text{ hours}$$

And,

$$\text{MTTF} = \frac{1}{\lambda_{IBC}} = 132.37 \text{ years}$$

From equation (11), reliability is determined as,

$$R_{IBC}(t) = 0.992 \quad (19)$$

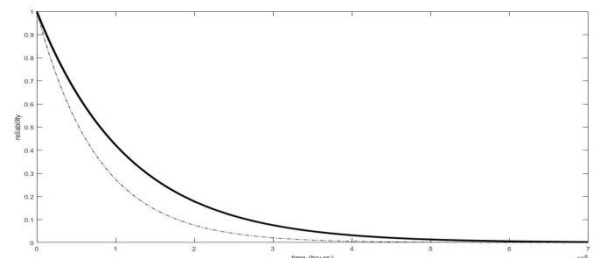


Fig. 8

Figure 8 shows the reliability plot for both conventional boost converter and IBC. The solid line plot is for IBC considered and dashed line plot is for normal boost converter.

VI. SIMULATION RESULTS

By using MATLAB/SIMULINK the simulation for IBC is designed. Characteristics of various parameters are depicted below.

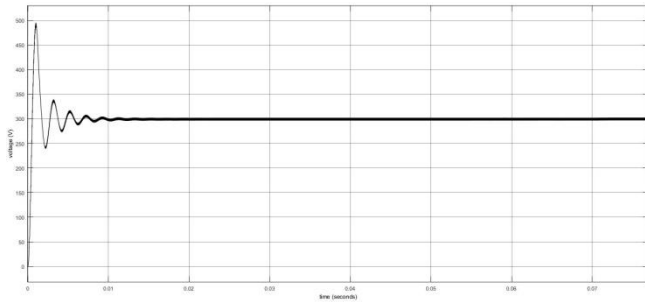


Fig.9

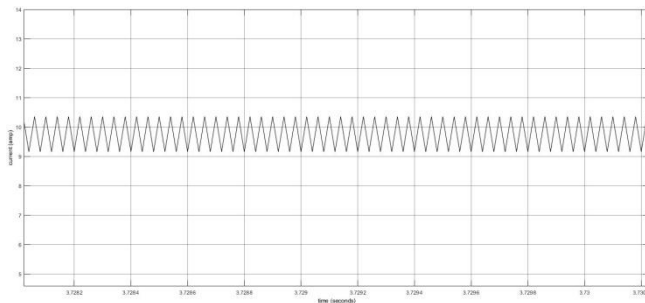


Fig.10

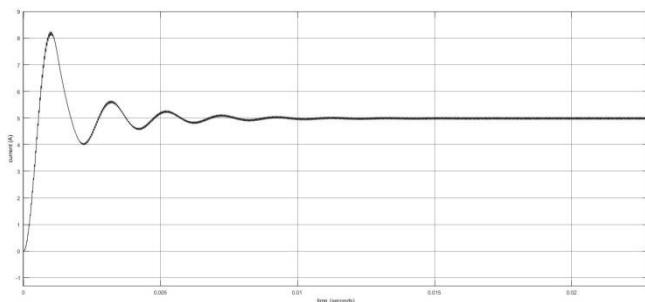


Fig.11

Figures (9,10,11) depict the output voltage characteristic, inductor current characteristic and output current characteristic respectively of IBC where output voltage is 300V, inductor current is 5.2A and output current is 5.03A.

VII. CONCLUSION AND FUTURE SCOPE

The modeling of Interleaved boost converter is discussed stepwise along with its simulation results with the help of MATLAB/SIMULINK. The failure rates of each component of IBC and the whole IBC are determined. The RBD model

is developed for a conventional boost converter and IBC and those are a series system and a parallel system respectively. With the help of this RBD the overall reliability evaluation and MTTF calculation are done for the IBC used in grid connected PV system. The interleaved boost converter acts as a power converter and MPP tracker as well because of its high reliable nature. The reliability evaluation can be done for the other topologies of these converters which are implemented in various other power generation system.

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